StudyForge

Cambridge IGCSE Combined Sciences

C5 Chemical Energetics

Pages 344-352

Comprehensive Exam Preparation Guide

C5 Chemical Energetics — Comprehensive Study Guide

Cambridge IGCSE Combined and Co-ordinated Sciences (Second Edition) | Pages 344–352

1. Chapter Overview

This chapter covers **chemical energetics** — the study of energy changes that happen during chemical reactions and physical changes. It is centred on one main topic:

- **C5.01 Exothermic and endothermic reactions** — Understanding that all reactions involve energy changes, classifying them as exothermic or endothermic, reading and drawing energy level diagrams, the concept of enthalpy change (ΔH), bond breaking vs bond making, and the role of activation energy.

Chemical energetics is a high-value exam topic because it connects to every other area of chemistry. You must be able to identify whether a reaction is exothermic or endothermic, draw and interpret energy level diagrams with correct labels (reactants, products, ΔH, activation energy E_a), explain the energy changes during bond breaking and bond making, and use these ideas to justify whether overall reactions release or absorb energy.

2. Key Concepts

2.1 Energy Changes in Chemical Reactions

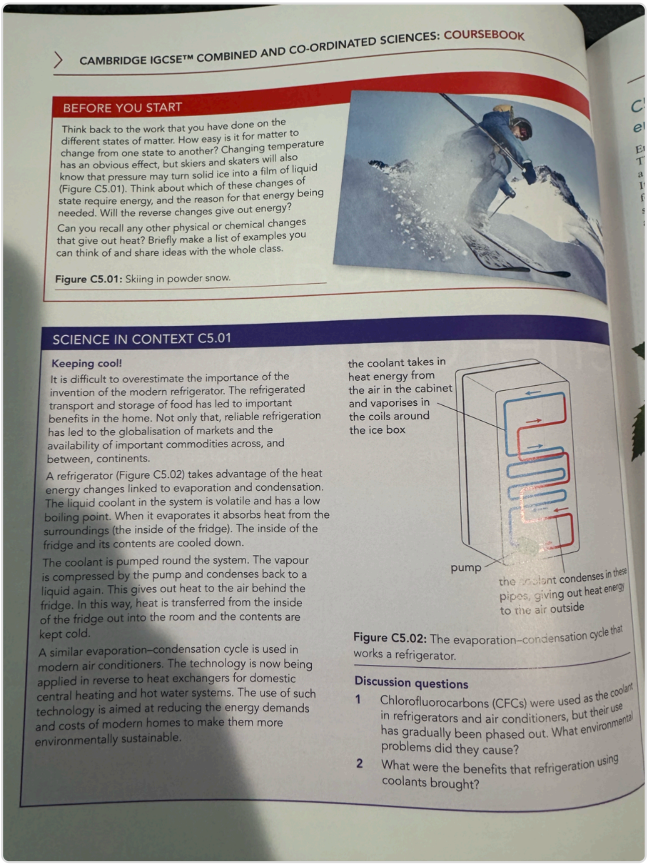
Every chemical reaction involves a transfer of thermal energy (heat) between the reaction system and its surroundings. When you carry out a reaction:

- **The system** = the reacting chemicals themselves
- **The surroundings** = everything else — the beaker, the water, the air, the thermometer, your hand

There are two possible directions of energy transfer:

1. Energy flows **out of** the system **into** the surroundings → **exothermic**
2. Energy flows **into** the system **from** the surroundings → **endothermic**

Key point for exams: You MUST describe exothermic and endothermic in terms of energy transfer to/from the surroundings, not just "gives out heat" or "takes in heat".



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BEFORE YOU START

Think back to the work that you have done on the different states of matter. How easy is it for matter to change from one state to another? Changing temperature has an obvious effect, but skiers and skaters will also know that pressure may turn solid ice into a film of liquid (Figure C5.01). Think about which of these changes of state require energy, and the reason for that energy being needed. Will the reverse changes give out energy?

Can you recall any other physical or chemical changes that give out heat? Briefly make a list of examples you can think of and share ideas with the whole class.

Figure C5.01: Skiing in powder snow.

SCIENCE IN CONTEXT C5.01

Keeping cool!

It is difficult to overestimate the importance of the invention of the modern refrigerator. The refrigerated transport and storage of food has led to important benefits in the home. Not only that, reliable refrigeration has led to the globalisation of markets and the availability of important commodities across, and between, continents.

A refrigerator (Figure C5.02) takes advantage of the heat energy changes linked to evaporation and condensation. The liquid coolant in the system is volatile and has a low boiling point. When it evaporates it absorbs heat from the surroundings (the inside of the fridge). The inside of the fridge and its contents are cooled down.

The coolant is pumped round the system. The vapour is compressed by the pump and condenses back to a liquid again. This gives out heat to the air behind the fridge. In this way, heat is transferred from the inside of the fridge out into the room and the contents are kept cold.

A similar evaporation–condensation cycle is used in modern air conditioners. The technology is now being applied in reverse to heat exchangers for domestic central heating and hot water systems. The use of such technology is aimed at reducing the energy demands and costs of modern homes to make them more environmentally sustainable.

the coolant takes in heat energy from the air in the cabinet and vaporises in the coils around the ice box

pump

the coolant condenses in these pipes, giving out heat energy to the air outside

Figure C5.02: The evaporation–condensation cycle that works a refrigerator.

Discussion questions

1 Chlorofluorocarbons (CFCs) were used as the coolant in refrigerators and air conditioners, but their use has gradually been phased out. What environmental problems did they cause?

2 What were the benefits that refrigeration using coolants brought?

Figure C5.01: Skiing in powder snow — energy from the surroundings changes physical states.

2.2 Exothermic Reactions

> **Exothermic reaction**: A reaction that **transfers thermal energy to the surroundings**, causing an **increase** in the temperature of the surroundings.

Memory aid: "**EX**othermic = energy **EX**its the reaction"

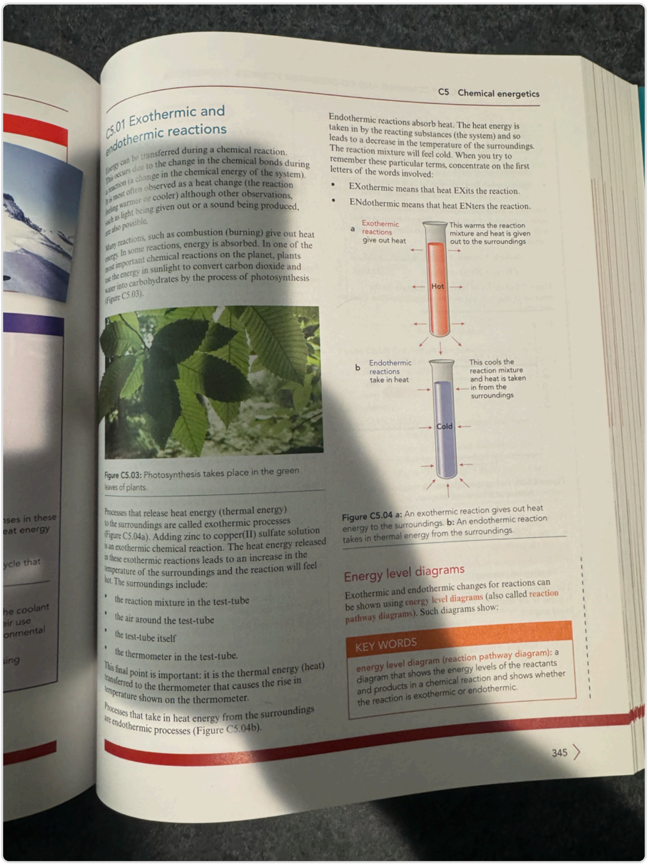
The surroundings get **hotter**. If you hold the beaker, it feels warm/hot. A thermometer placed in the reaction mixture shows a **temperature rise**.

Common examples of exothermic reactions:

- **Combustion** (burning fuels) — burning methane, wood, petrol: $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$
- **Neutralisation** — acid + alkali → salt + water
- **Oxidation reactions** — rusting of iron, respiration
- **Displacement reactions** — adding zinc to copper(II) sulfate solution: $Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$

Everyday exothermic processes:

- Hand warmers (crystallisation of sodium acetate)
- Self-heating food cans
- Setting of concrete/cement



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C5.01 Exothermic and endothermic reactions

Energy can be transferred during a chemical reaction. This occurs due to the change in the chemical bonds during a reaction (a change in the chemical energy of the system). It is most often observed as a heat change (the reaction feeling warmer or cooler) although other observations, such as light being given out or a sound being produced, are also possible.

Many reactions, such as combustion (burning) give out heat energy. In some reactions, energy is absorbed. In one of the most important chemical reactions on the planet, plants use the energy in sunlight to convert carbon dioxide and water into carbohydrates by the process of photosynthesis (Figure C5.03).

Figure C5.03: Photosynthesis takes place in the green leaves of plants.

Processes that release heat energy (thermal energy) to the surroundings are called exothermic processes (Figure C5.04a). Adding zinc to copper(II) sulfate solution is an exothermic chemical reaction. The heat energy released in these exothermic reactions leads to an increase in the temperature of the surroundings and the reaction will feel hot. The surroundings include:

- the reaction mixture in the test-tube
- the air around the test-tube
- the test-tube itself
- the thermometer in the test-tube.

This final point is important: it is the thermal energy (heat) transferred to the thermometer that causes the rise in temperature shown on the thermometer.

Processes that take in heat energy from the surroundings are endothermic processes (Figure C5.04b).

Endothermic reactions absorb heat. The heat energy is taken in by the reacting substances (the system) and so leads to a decrease in the temperature of the surroundings. The reaction mixture will feel cold. When you try to remember these particular terms, concentrate on the first letters of the words involved:

- EXothermic means that heat EXits the reaction.
- ENdothermic means that heat ENters the reaction.

Figure C5.04 a: An exothermic reaction gives out heat energy to the surroundings. b: An endothermic reaction takes in thermal energy from the surroundings.

Energy level diagrams

Exothermic and endothermic changes for reactions can be shown using energy level diagrams (also called reaction pathway diagrams). Such diagrams show:

KEY WORDS

energy level diagram (reaction pathway diagram): a diagram that shows the energy levels of the reactants and products in a chemical reaction and shows whether the reaction is exothermic or endothermic.

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Figure C5.03: Photosynthesis absorbs energy — it is endothermic. An exothermic reaction gives out heat — test tube feels warm.

2.3 Endothermic Reactions

> **Endothermic reaction**: A reaction that **takes in thermal energy from the surroundings**, causing a **decrease** in the temperature of the surroundings.

Memory aid: "**EN**dothermic = energy **EN**ters the reaction"

The surroundings get **colder**. If you hold the beaker, it feels cold. A thermometer shows a **temperature drop**.

Common examples of endothermic reactions:

- **Photosynthesis** — $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ (plants absorb light energy)
- **Thermal decomposition** — breaking down compounds with heat, e.g., $CaCO_3 \rightarrow CaO + CO_2$
- **Dissolving ammonium nitrate** in water — used in cold packs
- **Citric acid + sodium hydrogencarbonate** — temperature drops noticeably

Everyday endothermic processes:

- Cold packs for sports injuries
- Cooking (thermal decomposition of food)

2.4 Identifying Exothermic and Endothermic Reactions Experimentally

You can determine whether a reaction is exothermic or endothermic by measuring temperature changes:

Observation	Type of Reaction
Temperature of surroundings **increases**	Exothermic
Temperature of surroundings **decreases**	Endothermic

Beaker/container feels **hot**	Exothermic
Beaker/container feels **cold**	Endothermic
Thermometer reading **goes up**	Exothermic
Thermometer reading **goes down**	Endothermic

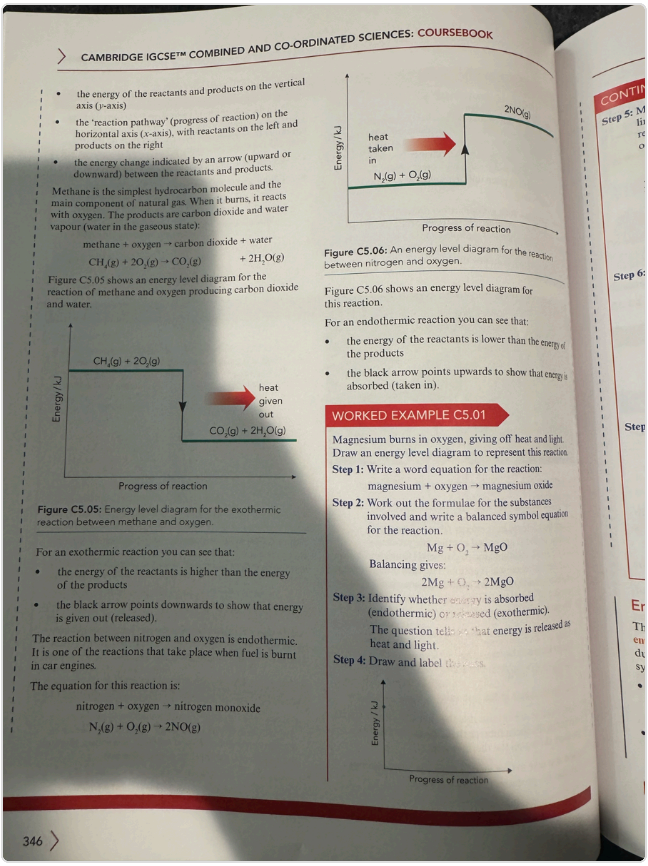
Exam tip: The question may describe a reaction where the temperature of the reaction mixture increases — this means the reaction is transferring heat TO the mixture (exothermic), causing it to warm up. The surroundings of the specific chemicals gain this thermal energy.

2.5 Energy Level Diagrams (Reaction Pathway Diagrams)

An **energy level diagram** (also called a **reaction pathway diagram**) is a graph that shows the energy levels of reactants and products in a chemical reaction, indicating whether the reaction is exothermic or endothermic.

How to read an energy level diagram:

- **Horizontal axis** = Progress of reaction (from reactants on the left to products on the right)
- **Vertical axis** = Energy (in kJ or kJ/mol)
- The **arrow** between reactant and product levels shows the **direction of energy change**:
- Arrow pointing **downwards** → **exothermic** (products have LESS energy than reactants)
- Arrow pointing **upwards** → **endothermic** (products have MORE energy than reactants)



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- the energy of the reactants and products on the vertical axis (*y*-axis)
- the 'reaction pathway' (progress of reaction) on the horizontal axis (*x*-axis), with reactants on the left and products on the right
- the energy change indicated by an arrow (upward or downward) between the reactants and products.

Methane is the simplest hydrocarbon molecule and the main component of natural gas. When it burns, it reacts with oxygen. The products are carbon dioxide and water vapour (water in the gaseous state):

methane + oxygen → carbon dioxide + water

$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$

Figure C5.05 shows an energy level diagram for the reaction of methane and oxygen producing carbon dioxide and water.

Figure C5.05: Energy level diagram for the exothermic reaction between methane and oxygen.

For an exothermic reaction you can see that:

- the energy of the reactants is higher than the energy of the products
- the black arrow points downwards to show that energy is given out (released).

The reaction between nitrogen and oxygen is endothermic. It is one of the reactions that take place when fuel is burnt in car engines.

The equation for this reaction is:

nitrogen + oxygen → nitrogen monoxide

$N_2(g) + O_2(g) \rightarrow 2NO(g)$

Figure C5.06: An energy level diagram for the reaction between nitrogen and oxygen.

Figure C5.06 shows an energy level diagram for this reaction.

For an endothermic reaction you can see that:

- the energy of the reactants is lower than the energy of the products
- the black arrow points upwards to show that energy is absorbed (taken in).

WORKED EXAMPLE C5.01

Magnesium burns in oxygen, giving off heat and light. Draw an energy level diagram to represent this reaction.

Step 1: Write a word equation for the reaction:

magnesium + oxygen → magnesium oxide

Step 2: Work out the formulae for the substances involved and write a balanced symbol equation for the reaction.

$Mg + O_2 \rightarrow MgO$

Balancing gives:

$2Mg + O_2 \rightarrow 2MgO$

Step 3: Identify whether energy is absorbed (endothermic) or released (exothermic).

The question tells us that energy is released as heat and light.

Step 4: Draw and label [illegible]

Figure C5.06: Energy level diagrams for exothermic (methane combustion) and endothermic (nitrogen + oxygen) reactions.

Exothermic Energy Level Diagram

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Energy

↑

| ____________

	Reactants
	____________

| |

| | ΔH (negative, arrow points DOWN)

| ↓

| ____________

	Products
	____________

|

└──────────────────────────→ Progress of reaction

`

- Reactants are at a **higher** energy level
- Products are at a **lower** energy level
- The difference = energy released to the surroundings
- ΔH is **negative** (e.g., ΔH = −728 kJ/mol for methane combustion)

Endothermic Energy Level Diagram

`

Energy

↑

| ____________

	Products
	____________

| ↑

| | ΔH (positive, arrow points UP)

| |

| ____________

	Reactants
	____________

|

└──────────────────────────→ Progress of reaction

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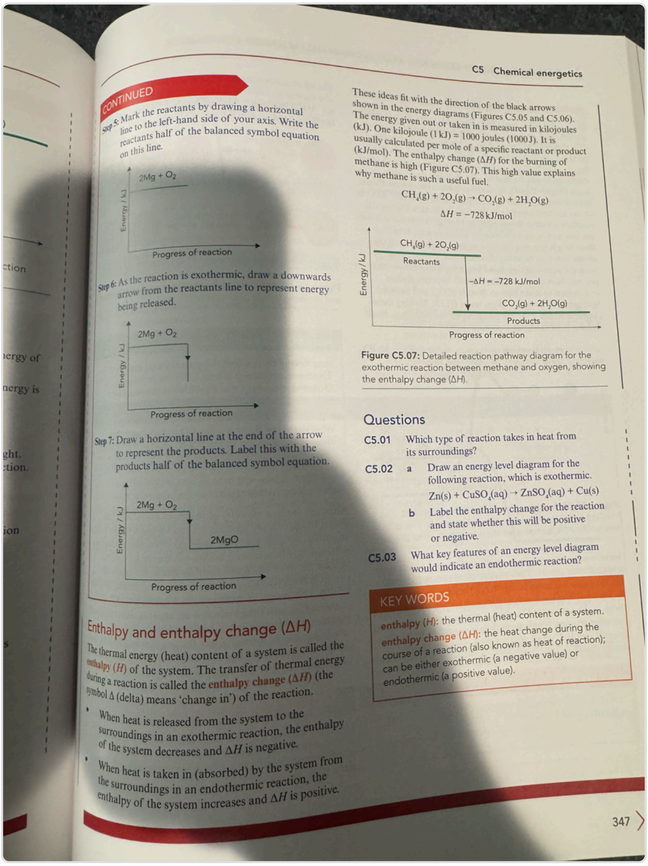
- Reactants are at a **lower** energy level
- Products are at a **higher** energy level
- The difference = energy absorbed from the surroundings
- ΔH is **positive**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Step 5: Mark the reactants by drawing a horizontal line to the left-hand side of your axis. Write the reactants half of the balanced symbol equation on this line.

2Mg + O_2
Energy / kJ
Progress of reaction

Step 6: As the reaction is exothermic, draw a downwards arrow from the reactants line to represent energy being released.

2Mg + O_2
Energy / kJ
Progress of reaction

Step 7: Draw a horizontal line at the end of the arrow to represent the products. Label this with the products half of the balanced symbol equation.

2Mg + O_2
2MgO
Energy / kJ
Progress of reaction

Enthalpy and enthalpy change (ΔH)

The thermal energy (heat) content of a system is called the enthalpy (H) of the system. The transfer of thermal energy during a reaction is called the enthalpy change (ΔH) (the symbol Δ (delta) means 'change in') of the reaction.

- When heat is released from the system to the surroundings in an exothermic reaction, the enthalpy of the system decreases and ΔH is negative.
- When heat is taken in (absorbed) by the system from the surroundings in an endothermic reaction, the enthalpy of the system increases and ΔH is positive.

These ideas fit with the direction of the black arrows shown in the energy diagrams (Figures C5.05 and C5.06). The energy given out or taken in is measured in kilojoules (kJ). One kilojoule (1 kJ) = 1000 joules (1000 J). It is usually calculated per mole of a specific reactant or product (kJ/mol). The enthalpy change (ΔH) for the burning of methane is high (Figure C5.07). This high value explains why methane is such a useful fuel.

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$$
$$\Delta H = -728\ kJ/mol$$

$CH_4(g) + 2O_2(g)$
Reactants
$-\Delta H = -728$ kJ/mol
$CO_2(g) + 2H_2O(g)$
Products
Energy / kJ
Progress of reaction

Figure C5.07: Detailed reaction pathway diagram for the exothermic reaction between methane and oxygen, showing the enthalpy change (ΔH).

Questions

C5.01 Which type of reaction takes in heat from its surroundings?

C5.02 a Draw an energy level diagram for the following reaction, which is exothermic.
$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$
b Label the enthalpy change for the reaction and state whether this will be positive or negative.

C5.03 What key features of an energy level diagram would indicate an endothermic reaction?

KEY WORDS

enthalpy (H): the thermal (heat) content of a system.
enthalpy change (ΔH): the heat change during the course of a reaction (also known as heat of reaction); can be either exothermic (a negative value) or endothermic (a positive value).

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Figure C5.07: Detailed energy level diagram for methane + oxygen showing enthalpy change.

2.6 Enthalpy and Enthalpy Change (ΔH)

> **Enthalpy (H)**: The total thermal energy (heat) content of a chemical system.

> **Enthalpy change (ΔH)**: The transfer of thermal energy during a reaction. Also known as the **heat of reaction**.

The symbol **Δ** (Greek letter delta) means "change in". So **ΔH = change in enthalpy**.

Reaction Type	Energy Transfer	ΔH Sign	Temperature of Surroundings
Exothermic	Energy released to surroundings	**Negative** (−)	Increases
Endothermic	Energy absorbed from surroundings	**Positive** (+)	Decreases

Worked Example — Methane combustion:

$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$ — ΔH = −728 kJ/mol

The negative sign tells us this is **exothermic** — 728 kJ of energy is released per mole of methane burned.

Exam tip: When asked to state whether a reaction is exothermic or endothermic from ΔH, look at the sign: negative = exothermic, positive = endothermic. You must always include the sign when writing ΔH values.

2.7 Drawing Energy Level Diagrams — Step by Step

Follow this method in exams:

Step 1: Write a word equation or balanced symbol equation for the reaction

Step 2: Determine whether the reaction is exothermic or endothermic

Step 3: Draw the diagram:

- Draw two horizontal lines — one for reactants, one for products
- For exothermic: products line is **below** reactants (energy released)
- For endothermic: products line is **above** reactants (energy absorbed)

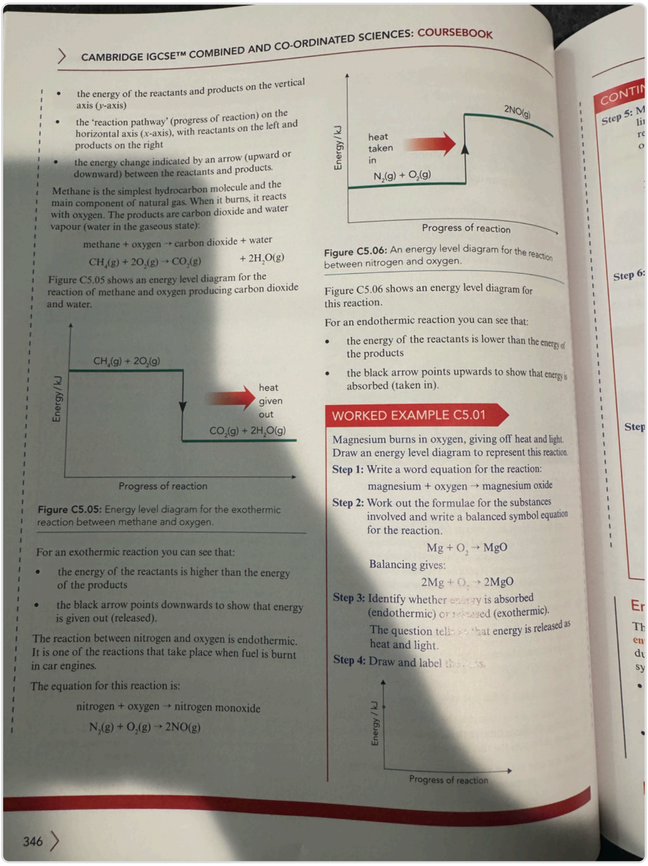
Step 4: Label everything:

- Label the reactants line with the formula(e)
- Label the products line with the formula(e)
- Draw an arrow showing ΔH (the energy change)
- Label the arrow with ΔH value and sign
- Label axes: "Energy / kJ" (vertical) and "Progress of reaction" (horizontal)

Worked Example C5.01 — Magnesium in oxygen:

$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$

This reaction gives off heat and light → **exothermic** → ΔH is negative → products are drawn **below** reactants.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- the energy of the reactants and products on the vertical axis (y-axis)
- the 'reaction pathway' (progress of reaction) on the horizontal axis (x-axis), with reactants on the left and products on the right
- the energy change indicated by an arrow (upward or downward) between the reactants and products.

Methane is the simplest hydrocarbon molecule and the main component of natural gas. When it burns, it reacts with oxygen. The products are carbon dioxide and water vapour (water in the gaseous state):

methane + oxygen → carbon dioxide + water

$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$

Figure C5.05 shows an energy level diagram for the reaction of methane and oxygen producing carbon dioxide and water.

Figure C5.05: Energy level diagram for the exothermic reaction between methane and oxygen.

For an exothermic reaction you can see that:

- the energy of the reactants is higher than the energy of the products
- the black arrow points downwards to show that energy is given out (released).

The reaction between nitrogen and oxygen is endothermic. It is one of the reactions that take place when fuel is burnt in car engines.

The equation for this reaction is:

nitrogen + oxygen → nitrogen monoxide

$N_2(g) + O_2(g) \rightarrow 2NO(g)$

Figure C5.06: An energy level diagram for the reaction between nitrogen and oxygen.

Figure C5.06 shows an energy level diagram for this reaction.

For an endothermic reaction you can see that:

- the energy of the reactants is lower than the energy of the products
- the black arrow points upwards to show that energy is absorbed (taken in).

WORKED EXAMPLE C5.01

Magnesium burns in oxygen, giving off heat and light. Draw an energy level diagram to represent this reaction.

Step 1: Write a word equation for the reaction:

magnesium + oxygen → magnesium oxide

Step 2: Work out the formulae for the substances involved and write a balanced symbol equation for the reaction.

$Mg + O_2 \rightarrow MgO$

Balancing gives:

$2Mg + O_2 \rightarrow 2MgO$

Step 3: Identify whether energy is absorbed (endothermic) or released (exothermic).

The question tells us that energy is released as heat and light.

Step 4: Draw and label the axes.

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Worked example: Drawing energy level diagram for magnesium burning in oxygen.

2.8 Bond Breaking and Bond Making

This is one of the most important concepts in the chapter. During every chemical reaction:

1. **Bonds in the reactants must be BROKEN** — this requires energy input → **endothermic** process
2. **New bonds in the products are MADE** — this releases energy → **exothermic** process

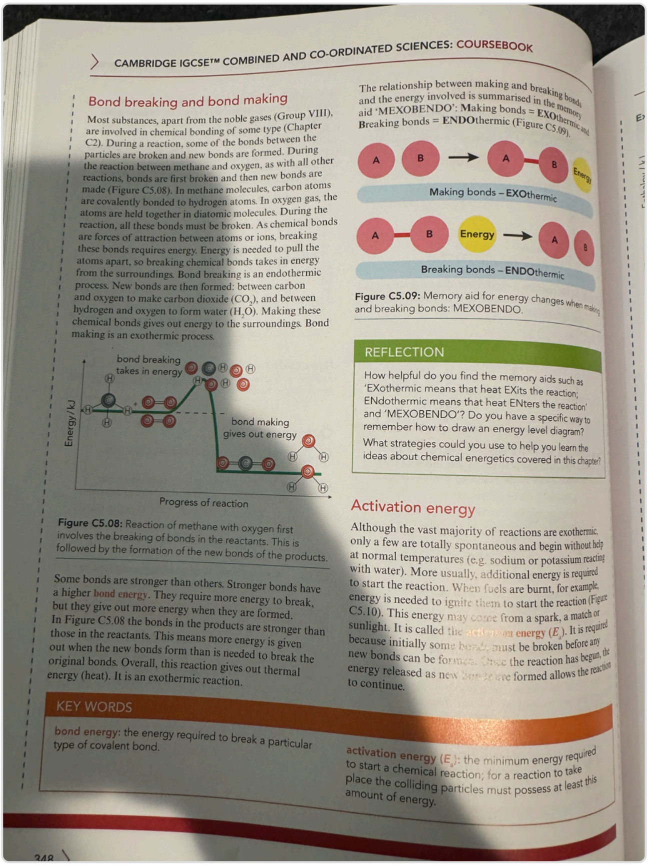
> **Bond breaking is ALWAYS endothermic** (energy must be put IN to pull atoms apart)

>

> **Bond making is ALWAYS exothermic** (energy is given OUT when atoms come together)

Memory aid: MEXOBENDO

- **M**aking bonds = **EXO**thermic
- **B**reaking bonds = **ENDO**thermic



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Bond breaking and bond making

Most substances, apart from the noble gases (Group VIII), are involved in chemical bonding of some type (Chapter C2). During a reaction, some of the bonds between the particles are broken and new bonds are formed. During the reaction between methane and oxygen, as with all other reactions, bonds are first broken and then new bonds are made (Figure C5.08). In methane molecules, carbon atoms are covalently bonded to hydrogen atoms. In oxygen gas, the atoms are held together in diatomic molecules. During the reaction, all these bonds must be broken. As chemical bonds are forces of attraction between atoms or ions, breaking these bonds requires energy. Energy is needed to pull the atoms apart, so breaking chemical bonds takes in energy from the surroundings. Bond breaking is an endothermic process. New bonds are then formed: between carbon and oxygen to make carbon dioxide (CO_2), and between hydrogen and oxygen to form water (H_2O). Making these chemical bonds gives out energy to the surroundings. Bond making is an exothermic process.

Figure C5.08: Reaction of methane with oxygen first involves the breaking of bonds in the reactants. This is followed by the formation of the new bonds of the products.

Some bonds are stronger than others. Stronger bonds have a higher bond energy. They require more energy to break, but they give out more energy when they are formed. In Figure C5.08 the bonds in the products are stronger than those in the reactants. This means more energy is given out when the new bonds form than is needed to break the original bonds. Overall, this reaction gives out thermal energy (heat). It is an exothermic reaction.

The relationship between making and breaking bonds and the energy involved is summarised in the memory aid 'MEXOBENDO': Making bonds = EXOthermic and Breaking bonds = ENDOthermic (Figure C5.09).

Figure C5.09: Memory aid for energy changes when making and breaking bonds: MEXOBENDO.

REFLECTION

How helpful do you find the memory aids such as 'EXothermic means that heat EXits the reaction; ENdothermic means that heat ENters the reaction' and 'MEXOBENDO'? Do you have a specific way to remember how to draw an energy level diagram?

What strategies could you use to help you learn the ideas about chemical energetics covered in this chapter?

Activation energy

Although the vast majority of reactions are exothermic, only a few are totally spontaneous and begin without help at normal temperatures (e.g. sodium or potassium reacting with water). More usually, additional energy is required to start the reaction. When fuels are burnt, for example, energy is needed to ignite them to start the reaction (Figure C5.10). This energy may come from a spark, a match or sunlight. It is called the activation energy (E_a). It is required because initially some bonds must be broken before any new bonds can be formed. Once the reaction has begun, the energy released as new bonds are formed allows the reaction to continue.

KEY WORDS

bond energy: the energy required to break a particular type of covalent bond.

activation energy (E_a): the minimum energy required to start a chemical reaction; for a reaction to take place the colliding particles must possess at least this amount of energy.

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Figure C5.08: Reaction of methane with oxygen — bond breaking absorbs energy, bond making releases energy.

How the Overall Reaction Type is Determined

Whether a reaction is overall exothermic or endothermic depends on the **balance** between energy absorbed (breaking bonds) and energy released (making bonds):

Scenario	Overall Reaction
Energy released by making new bonds **>** Energy absorbed by breaking old bonds	**Exothermic** (overall energy given out)
Energy released by making new bonds **<** Energy absorbed by breaking old bonds	**Endothermic** (overall energy taken in)

Example — Combustion of methane (exothermic):

- Breaking 4 C−H bonds and 2 O=O bonds requires energy (endothermic step)
- Making 2 C=O bonds and 4 O−H bonds releases energy (exothermic step)
- The energy released by making the new bonds is **greater** than the energy absorbed by breaking the old bonds
- Net result: energy is released → exothermic overall

Bond Energy

> **Bond energy (E)**: The energy required to break a particular type of covalent bond. The same amount of energy is released when that bond is formed.

- Stronger bonds have **higher** bond energies (more energy needed to break them, more energy released when they form)
- Different types of bonds have different bond energies (e.g., C=O is stronger than C−H)

2.9 Activation Energy

> **Activation energy (E_a)**: The **minimum** amount of energy that colliding particles must possess for a reaction to take place.

Even exothermic reactions that release large amounts of energy need a small input of energy to get started. This is the activation energy.

Everyday examples of activation energy:

- Striking a **match** — the friction provides the activation energy to start the combustion
- A **spark** in a car engine — ignites the fuel-air mixture
- **Lighting** a Bunsen burner — a lighter provides the initial energy

Why do exothermic reactions need activation energy?

The reactant molecules must collide with **enough energy** to break the bonds in the reactants. Once these initial bonds are broken, the reaction proceeds and the energy released from making new bonds sustains the reaction.

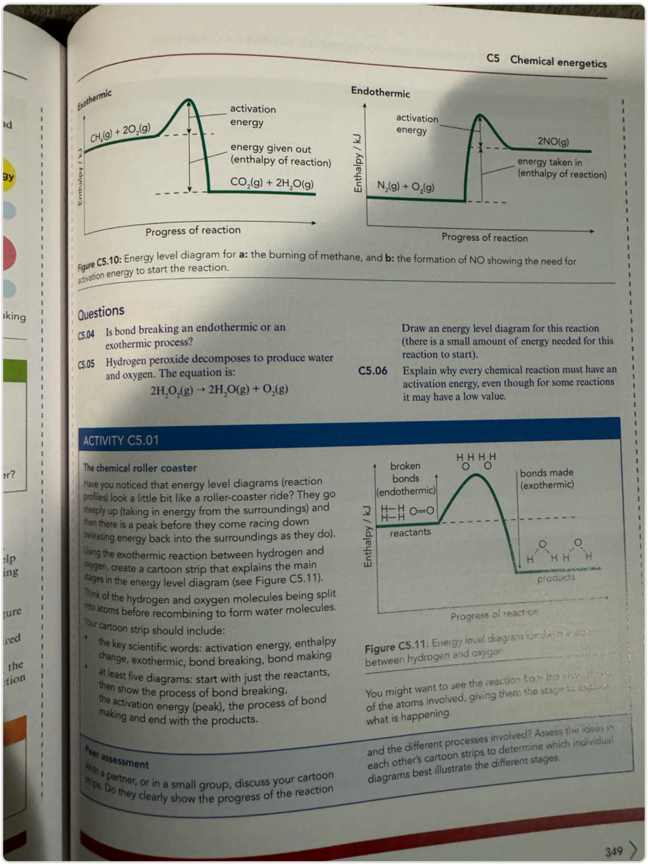
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Figure C5.10: Energy level diagram for **a:** the burning of methane, and **b:** the formation of NO showing the need for activation energy to start the reaction.

Questions

C5.04 Is bond breaking an endothermic or an exothermic process?

C5.05 Hydrogen peroxide decomposes to produce water and oxygen. The equation is:

$$2H_2O_2(g) \rightarrow 2H_2O(g) + O_2(g)$$

Draw an energy level diagram for this reaction (there is a small amount of energy needed for this reaction to start).

C5.06 Explain why every chemical reaction must have an activation energy, even though for some reactions it may have a low value.

ACTIVITY C5.01

The chemical roller coaster

Have you noticed that energy level diagrams (reaction profiles) look a little bit like a roller-coaster ride? They go steeply up (taking in energy from the surroundings) and then there is a peak before they come racing down (releasing energy back into the surroundings as they do).

Using the exothermic reaction between hydrogen and oxygen, create a cartoon strip that explains the main stages in the energy level diagram (see Figure C5.11).

Think of the hydrogen and oxygen molecules being split into atoms before recombining to form water molecules.

Your cartoon strip should include:

- the key scientific words: activation energy, enthalpy change, exothermic, bond breaking, bond making
- at least five diagrams: start with just the reactants, then show the process of bond breaking, the activation energy (peak), the process of bond making and end with the products.

Figure C5.11: Energy level diagram [illegible] between hydrogen and oxygen

You might want to see the reaction from the [illegible] of the atoms involved, giving them the stage to [illegible] what is happening.

Peer assessment

With a partner, or in a small group, discuss your cartoon strips. Do they clearly show the progress of the reaction and the different processes involved? Assess the ideas in each other's cartoon strips to determine which individual diagrams best illustrate the different stages.

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Figure C5.10: Energy level diagram showing activation energy for exothermic and endothermic reactions.

Energy Level Diagrams with Activation Energy

When activation energy is included, the energy level diagram shows a **"hill" or "hump"** that the reactants must overcome:

Exothermic reaction with activation energy:

`

Energy

↑ ╱╲

| ╱ ╲ ← activation energy (E_a)

| _____╱ ╲

| |Reactants| ╲

| |__________| ╲_____

	Products
	_________
	← ΔH →

└──────────────────────────────→ Progress of reaction

`

Endothermic reaction with activation energy:

`

Energy

↑ ╱╲

| ╱ ╲ ← activation energy (E_a)

| ╱ ╲_____

╱	Products
╱	_________

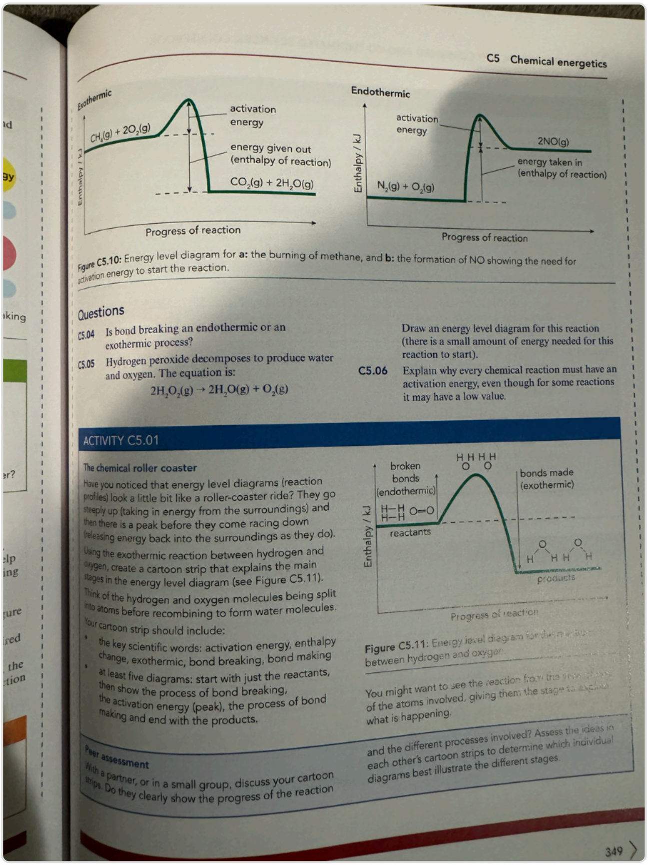
| _____╱

	Reactants
	_________

└──────────────────────────────→ Progress of reaction

`

Key exam point: Activation energy is measured from the **energy level of the reactants** to the **top of the energy hump** — NOT from the products level.



C5 Chemical energetics

Figure C5.10: Energy level diagram for **a:** the burning of methane, and **b:** the formation of NO showing the need for activation energy to start the reaction.

Questions

C5.04 Is bond breaking an endothermic or an exothermic process?

C5.05 Hydrogen peroxide decomposes to produce water and oxygen. The equation is:

$$2H_2O_2(g) \rightarrow 2H_2O(g) + O_2(g)$$

Draw an energy level diagram for this reaction (there is a small amount of energy needed for this reaction to start).

C5.06 Explain why every chemical reaction must have an activation energy, even though for some reactions it may have a low value.

ACTIVITY C5.01

The chemical roller coaster

Have you noticed that energy level diagrams (reaction profiles) look a little bit like a roller-coaster ride? They go steeply up (taking in energy from the surroundings) and then there is a peak before they come racing down (releasing energy back into the surroundings as they do).

Using the exothermic reaction between hydrogen and oxygen, create a cartoon strip that explains the main stages in the energy level diagram (see Figure C5.11). Think of the hydrogen and oxygen molecules being split into atoms before recombining to form water molecules.

Your cartoon strip should include:

- the key scientific words: activation energy, enthalpy change, exothermic, bond breaking, bond making
- at least five diagrams: start with just the reactants, then show the process of bond breaking, the activation energy (peak), the process of bond making and end with the products.

Figure C5.11: Energy level diagram ... between hydrogen and oxygen

You might want to see the reaction from the ... of the atoms involved, giving them the stage to ... what is happening.

Peer assessment

With a partner, or in a small group, discuss your cartoon strips. Do they clearly show the progress of the reaction and the different processes involved? Assess the ideas in each other's cartoon strips to determine which individual diagrams best illustrate the different stages.

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Figure C5.11: Energy level diagram for hydrogen and oxygen reaction with activation energy.

3. Important Terms and Definitions

Term	Definition
Exothermic reaction	A reaction that transfers thermal energy to the surroundings, causing an increase in temperature
Endothermic reaction	A reaction that takes in thermal energy from the surroundings, causing a decrease in temperature
Enthalpy (H)	The total thermal energy (heat) content of a chemical system
Enthalpy change (ΔH)	The transfer of thermal energy during a reaction; also called heat of reaction
Energy level diagram	A diagram showing the energy levels of reactants and products, indicating whether a reaction is exothermic or endothermic
Bond energy (E)	The energy required to break one particular type of covalent bond; the same energy is released when that bond forms
Activation energy (E_a)	The minimum amount of energy that colliding particles must possess for a reaction to occur
Bond breaking	The process of pulling atoms apart by overcoming attractive forces; always endothermic
Bond making	The process of atoms coming together to form new bonds; always exothermic
System	The chemicals involved in the reaction
Surroundings	Everything outside the system — the container, air, water, thermometer

4. Key Equations and Energy Changes

Chemical Equations with Energy Information

Reaction	Equation	Type	ΔH
Combustion of methane	$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$	Exothermic	−728 kJ/mol
Magnesium combustion	$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$	Exothermic	Negative
Nitrogen + oxygen	$N_2(g) + O_2(g) \rightarrow 2NO(g)$	Endothermic	Positive
Photosynthesis	$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$	Endothermic	Positive
Zinc + copper sulfate	$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$	Exothermic	Negative
Decomposition of hydrogen peroxide	$2H_2O_2(l) \rightarrow 2H_2O(l) + O_2(g)$	Exothermic	Negative
Sodium hydrogencarbonate + HCl	$NaHCO_3 + HCl \rightarrow NaCl + CO_2 + H_2O$	Endothermic	Positive
Nitrogen + hydrogen (ammonia)	$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$	Exothermic	Negative

Key Relationships

- **ΔH negative** → exothermic → energy released → products lower than reactants
- **ΔH positive** → endothermic → energy absorbed → products higher than reactants
- **Overall exothermic:** Energy released from bond making > Energy absorbed by bond breaking
- **Overall endothermic:** Energy released from bond making < Energy absorbed by bond breaking

5. Exam Focus Points

What Examiners Look For

1. **Define exothermic and endothermic correctly** — always refer to "transfer of thermal energy to/from the surroundings" and the "increase/decrease in temperature of the surroundings." Do NOT just say "gives out/takes in heat."

1. **Draw energy level diagrams accurately:**

- Label BOTH axes (Energy / kJ on vertical, Progress of reaction on horizontal)
- Label reactants and products with their formulae
- Show ΔH with an arrow and correct direction
- Include activation energy E_a as a hump when asked
- Ensure E_a is measured from reactant level to the peak, NOT from product level

1. **State the sign of ΔH:**

- Exothermic = **negative ΔH**
- Endothermic = **positive ΔH**
- Always include the sign when writing ΔH values

1. **Bond breaking and making:**

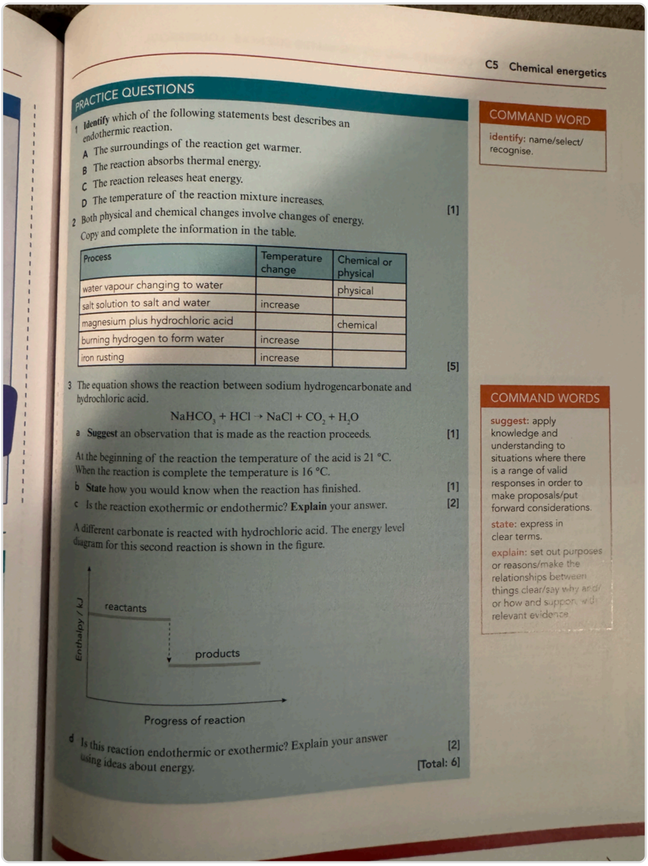
- Bond breaking = endothermic (energy absorbed)
- Bond making = exothermic (energy released)
- Overall reaction type depends on which process releases/absorbs more energy

1. **Activation energy questions:**

- Define it as the MINIMUM energy for a reaction to start
- Explain that all reactions need activation energy, even exothermic ones
- On diagrams, E_a is the height from reactants to the top of the energy "hump"

Common Exam Mistakes to Avoid

- Saying "the reaction gets hot" instead of "the surroundings temperature increases"
- Drawing ΔH from the wrong level (it goes from reactants to products, not from the peak)
- Confusing E_a with ΔH — they are different quantities
- Forgetting that bond breaking in BOTH exothermic and endothermic reactions is endothermic
- Writing ΔH without the correct sign (+ or −)
- Drawing energy level diagrams without axis labels


C5 Chemical energetics

PRACTICE QUESTIONS

1 **Identify** which of the following statements best describes an endothermic reaction.

A The surroundings of the reaction get warmer.
B The reaction absorbs thermal energy.
C The reaction releases heat energy.
D The temperature of the reaction mixture increases. [1]

2 Both physical and chemical changes involve changes of energy. Copy and complete the information in the table.

Process	Temperature change	Chemical or physical
water vapour changing to water		physical
salt solution to salt and water	increase	
magnesium plus hydrochloric acid		chemical
burning hydrogen to form water	increase	
iron rusting	increase	

[5]

3 The equation shows the reaction between sodium hydrogencarbonate and hydrochloric acid.

$$NaHCO_3 + HCl \rightarrow NaCl + CO_2 + H_2O$$

a **Suggest** an observation that is made as the reaction proceeds. [1]

At the beginning of the reaction the temperature of the acid is 21 °C. When the reaction is complete the temperature is 16 °C.

b **State** how you would know when the reaction has finished. [1]
c Is the reaction exothermic or endothermic? **Explain** your answer. [2]

A different carbonate is reacted with hydrochloric acid. The energy level diagram for this second reaction is shown in the figure.

d Is this reaction endothermic or exothermic? Explain your answer using ideas about energy. [2]
[Total: 6]

COMMAND WORD
identify: name/select/recognise.

COMMAND WORDS
suggest: apply knowledge and understanding to situations where there is a range of valid responses in order to make proposals/put forward considerations.
state: express in clear terms.
explain: set out purposes or reasons/make the relationships between things clear/say why and/or how and support with relevant evidence

Practice questions and command words for C5 Chemical Energetics.

6. Practice Questions with Full Answers

Question C5.01

Which type of reaction takes in heat from its surroundings?

Answer: An endothermic reaction takes in thermal energy from the surroundings, causing a decrease in the temperature of the surroundings.

Question C5.02 (4 marks)

a) Draw an energy level diagram for the following reaction, which is exothermic: Zn(s) + $CuSO_4$(aq) → $ZnSO_4$(aq) + Cu(s). b) Label the enthalpy change for the reaction and state whether this will be positive or negative.

Answer: The energy level diagram should show: reactants (Zn + $CuSO_4$) at a higher energy level, products ($ZnSO_4$ + Cu) at a lower energy level, a downward arrow labelled ΔH between the two levels. The x-axis is labelled "Progress of reaction" and the y-axis is labelled "Energy / kJ". The enthalpy change ΔH is negative because the reaction is exothermic — energy is released to the surroundings.

Question C5.03 (3 marks)

What key features of an energy level diagram would indicate an endothermic reaction?

Answer: Three features indicate an endothermic reaction: (1) The products are shown at a higher energy level than the reactants. (2) The arrow representing ΔH points upwards from reactants to products. (3) The value of ΔH is positive, showing that energy has been absorbed from the surroundings.

Question C5.04

Is bond breaking an endothermic or an exothermic process? Explain your answer.

Answer: Bond breaking is an endothermic process. Energy must be supplied (absorbed from the surroundings) to overcome the attractive forces holding atoms together in a bond. This means energy enters the system when bonds are broken.

Question C5.05 (3 marks)

Hydrogen peroxide decomposes to produce water and oxygen. The equation is: $2H_2O_2(l) \rightarrow 2H_2O(l) + O_2(g)$. Explain why every chemical reaction must have an activation energy even though for some reactions it may have a low value.

Answer: Every chemical reaction requires activation energy because bonds in the reactant molecules must first be broken before new bonds can form. Breaking bonds is always an endothermic process that requires energy input. The colliding particles must possess at least a minimum amount of energy (the activation energy) to break these existing bonds and allow the reaction to proceed, even if the overall reaction releases energy (is exothermic).

Question C5.06 (2 marks)

Define the term enthalpy change (ΔH).

Answer: The enthalpy change (ΔH) is the transfer of thermal energy during a chemical reaction. It is also known as the heat of reaction. For exothermic reactions ΔH is negative; for endothermic reactions ΔH is positive.

Question C5.07 (3 marks)

Methane burns in oxygen according to: $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$, ΔH = −728 kJ/mol. Explain what the negative sign of ΔH tells you about this reaction.

Answer: The negative sign of ΔH tells us the reaction is exothermic. This means that thermal energy is transferred from the reacting chemicals (the system) to the surroundings. The products (CO_2 and H_2O) have less energy than the reactants (CH_4 and O_2), and 728 kJ of energy is released per mole of methane burned.

Question C5.08 (4 marks)

Explain why the combustion of methane is exothermic in terms of bond breaking and bond making.

Answer: During the combustion of methane, bonds in the reactants must first be broken: four C−H bonds in methane and two O=O bonds in oxygen molecules. This step absorbs energy (endothermic).

New bonds are then formed in the products: two C=O bonds in carbon dioxide and four O−H bonds in water. This step releases energy (exothermic). The total energy released by making the new bonds is greater than the total energy absorbed by breaking the old bonds, so the overall reaction is exothermic — there is a net release of energy to the surroundings.

Question C5.09 (2 marks)

State what is meant by the term activation energy.

Answer: Activation energy (E_a) is the minimum amount of energy that colliding particles must possess in order for a chemical reaction to take place. It is the energy needed to start breaking bonds in the reactant molecules.

Question C5.10 (5 marks)

The equation for the reaction between sodium hydrogencarbonate and hydrochloric acid is: $NaHCO_3 + HCl \rightarrow NaCl + CO_2 + H_2O$. At the beginning of the reaction the temperature of the acid is 21 °C. When the reaction is complete the temperature is 16 °C. a) Suggest an observation as the reaction proceeds. b) State how you would know when the reaction has finished. c) Is this reaction exothermic or endothermic? Explain your answer.

Answer: a) Bubbles (effervescence/fizzing) would be observed as carbon dioxide gas is produced. b) The reaction has finished when the bubbling/effervescence stops and no more gas is being produced. c) This reaction is endothermic. The temperature of the surroundings (the solution) decreased from 21 °C to 16 °C. This shows that thermal energy was absorbed from the surroundings by the reacting chemicals, causing the surroundings to cool down. The enthalpy change ΔH for this reaction is positive.

Question C5.11 (3 marks)

The reaction between nitrogen and oxygen in a car engine is endothermic: $N_2(g) + O_2(g) \rightarrow 2NO(g)$. On the energy level diagram for this reaction, describe where the reactants, products, ΔH, and activation energy (E_a) would be positioned.

Answer: On the energy level diagram: the reactants ($N_2 + O_2$) would be drawn at a lower energy level. The products (2NO) would be drawn at a higher energy level (because the reaction is endothermic). The ΔH arrow would point upwards from the reactants line to the products line, and would be labelled with a positive ΔH value. The activation energy (E_a) would be shown as a hump above the reactants level — measured from the reactants line to the top of the energy hump, not from the products line.

Question C5.12 (2 marks)

A student says: "Bond breaking is exothermic because you need to heat the substance to break the bonds, so heat exits." Explain why the student is wrong.

Answer: The student is incorrect. Bond breaking is endothermic, not exothermic. When bonds are broken, energy is absorbed from the surroundings into the system to overcome the attractive forces between atoms. "Endothermic" means energy enters the system. The fact that you need to supply heat actually proves it is endothermic — the energy goes INTO the bond-breaking process, not out of it.

Practice Question 1 (1 mark)

Identify which of the following statements best describes an exothermic reaction: A. The surroundings of the reaction get warmer. B. The reaction absorbs thermal energy. C. The temperature of the reaction mixture increases. D. The reaction releases heat energy.

Answer: The correct answer is A — "The surroundings of the reaction get warmer." This is the precise definition: an exothermic reaction transfers thermal energy TO the surroundings, causing their temperature to increase. Option D is imprecise and option C refers to the mixture, not the surroundings.

Practice Question 2 (5 marks)

Copy and complete the table for each process, stating whether the temperature change is an increase or decrease, and whether the change is chemical or physical: water vapour changing to water, salt solution to salt and water, magnesium plus hydrochloric acid, burning hydrogen to form water, iron rusting.

Answer: Water vapour → water: temperature increase, physical change (condensation is exothermic). Salt solution → salt and water: temperature increase, physical change (evaporation requires heat input, but crystallisation releases heat). Magnesium + hydrochloric acid: temperature increase, chemical change (displacement/neutralisation — exothermic). Burning hydrogen → water: temperature increase, chemical change (combustion — exothermic). Iron rusting: temperature increase, chemical change (oxidation — exothermic, though very slow).

Practice Question 3 (6 marks)

A different carbonate is reacted with hydrochloric acid. The energy level diagram shows products higher than reactants. a) Is this reaction exothermic or endothermic? Explain your answer. b) What happens to the temperature of the surroundings during this reaction? c) What is the sign of ΔH for this reaction?

Answer: a) This reaction is endothermic because the products are at a higher energy level than the reactants on the energy level diagram. This means the system has absorbed energy from the surroundings. b) The temperature of the surroundings decreases. Thermal energy is transferred from the surroundings into the system (the reacting chemicals), so the surroundings cool down. c) ΔH is positive for this reaction, because energy is absorbed from the surroundings (endothermic reactions always have a positive ΔH).

Practice Question 4 (2 marks)

Nitrogen reacts with hydrogen to form ammonia: $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$. The figure shows the energy pathway diagram for the reaction between hydrogen and nitrogen. On the energy profile, label the overall enthalpy change ΔH and the activation energy E_a.

Answer: ΔH should be labelled as the vertical distance between the energy level of the reactants (N_2 + $3H_2$) and the energy level of the products ($2NH_3$). Since the products are at a lower energy level, ΔH is negative (exothermic). E_a (activation energy) should be labelled as the vertical distance from the reactants' energy level to the top of the energy "hump" (the peak of the curve). E_a is always measured upwards from the reactants, not from the products.

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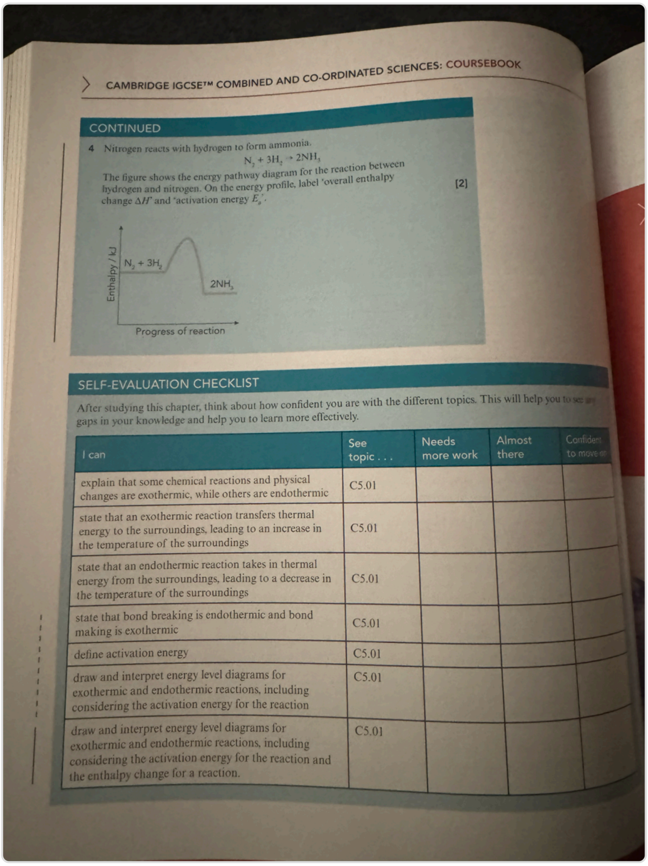
7. Quick Revision Card

5 things to remember for C5 Chemical Energetics:

1. **Exothermic** = energy exits → surroundings get warmer → ΔH is **negative** → products are LOWER on energy level diagram
1. **Endothermic** = energy enters → surroundings get cooler → ΔH is **positive** → products are HIGHER on energy level diagram
1. **MEXOBENDO**: **M**aking bonds = **EXO**thermic (releases energy) | **B**reaking bonds = **ENDO**thermic (absorbs energy)
1. **Activation energy (E_a)** = minimum energy needed to start a reaction. All reactions need it — even highly exothermic ones. On energy level diagrams, it's the "hill" above the reactant line.

1. **Overall reaction type** = compare total bond breaking energy vs total bond making energy. If making releases MORE → exothermic. If breaking absorbs MORE → endothermic.

Quick self-test: Can you draw an energy level diagram for the combustion of methane, label reactants, products, ΔH, and E_a, and explain why it is exothermic using bond energy ideas?



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CONTINUED

4 Nitrogen reacts with hydrogen to form ammonia.

$$N_2 + 3H_2 \rightarrow 2NH_3$$

The figure shows the energy pathway diagram for the reaction between hydrogen and nitrogen. On the energy profile, label 'overall enthalpy change ΔH' and 'activation energy E_a'. [2]

SELF-EVALUATION CHECKLIST

After studying this chapter, think about how confident you are with the different topics. This will help you to see any gaps in your knowledge and help you to learn more effectively.

I can	See topic ...	Needs more work	Almost there	Confident to move on
explain that some chemical reactions and physical changes are exothermic, while others are endothermic	C5.01			
state that an exothermic reaction transfers thermal energy to the surroundings, leading to an increase in the temperature of the surroundings	C5.01			
state that an endothermic reaction takes in thermal energy from the surroundings, leading to a decrease in the temperature of the surroundings	C5.01			
state that bond breaking is endothermic and bond making is exothermic	C5.01			
define activation energy	C5.01			
draw and interpret energy level diagrams for exothermic and endothermic reactions, including considering the activation energy for the reaction	C5.01			
draw and interpret energy level diagrams for exothermic and endothermic reactions, including considering the activation energy for the reaction and the enthalpy change for a reaction.	C5.01			

Self-evaluation checklist for C5 Chemical Energetics.